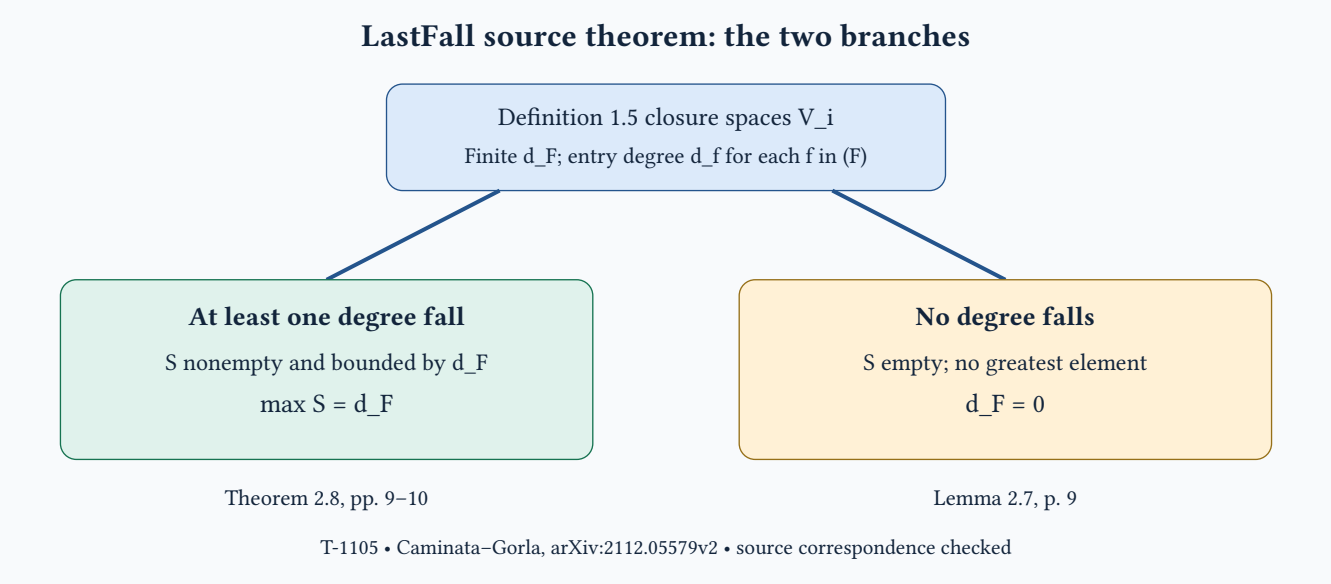
LastFall source theorem and hypotheses review — T-1105



Date: 2026-10-31. Reviewer: independent source session /root/lastfall_source.

Assigned-joint verdict: holds. The conditional mathematical claim in the frozen target faithfully states Caminata–Gorla Theorem 2.8. Its explicit finiteness assumption exposes an ingredient the paper establishes earlier. The hypothesis that a degree fall exists is essential to a greatest-element conclusion. The paper and target quantify the maximum over the generated ideal, rather than merely over the original generators.

This verdict covers the source theorem, Definition 1.5, and their hypotheses. It is not a machine verification or an evaluation of the finite-profile claim. No Lean source, build receipt, or sibling review was read; no Lean build ran. The coordinator owns the combined outcome.

Immutable inputs and primary-source access

- Review plan: commit 2b0cfe1401831c9493cfe047a515955f38d037b8, docs/reports/2026-10-31-lastfall-fidelity/review_plan.yaml; SHA-256 8326616dea35f3fa45f3957fa8650338598a7fcf9d42ff274afaedcb48cb156e.
- Claim snapshot and fetched origin/main: 199f3aaaa6d37d2fce59f4d8c8fe8d2eb8b50672.
- Target: formal/targets/lastfall-is-max-degree-fall.yaml, especially lines 27–31 and 41–73; SHA-256 74ecb7ee9f8dcf67e14e732bd8382498f648147a6a002d258e1d7b634e065ea9.
- Producer note: research/THM_FALLDEG_INVARIANTS1.md, especially lines 24–35; SHA-256 18a9ffd4c94288c097bb25d1908fbac1bd23348ed69dc3508b61ddbc5a9a4331.
- Primary source: Alessio Caminata and Elisa Gorla, Solving degree, last fall degree, and related invariants, ePrint 2021/1611 (https://eprint.iacr.org/2021/1611), final revision dated 2022-06-01. The ePrint landing page was read; PDF access returned HTTP 403 through both web and curl.
- The authors' corresponding arXiv:2112.05579v2 PDF (https://arxiv.org/pdf/2112.05579v2), dated 2022-06-01, was downloaded and read in full, including the references. SHA-256 9e9323db0cd2f755a375c1baa836b999972fc38eebca3e6f55f63dfecb226177. The matching title, authors, revision date, 16-page extent, and theorem numbering support use of this primary-source version; byte identity with the inaccessible ePrint PDF was not checked.

Paper citations below use the printed page number. Local extraction line numbers refer to pdftotext -layout output from that exact PDF: setting on p. 3, lines 128–142; Definition 1.5 and finiteness on p. 6, lines 245–264; Definitions 2.5–2.6 and Lemma 2.7 on p. 9, lines 373–388; Theorem 2.8 and proof on pp. 9–10, lines 392–406. Pages 6 and 9 were also visually inspected.

Source correspondence

Clause	Primary source	Frozen target	Assessment
Field and ring	p. 3, §1: arbitrary field; finitely many variables	lines 42–43	Matches; no finite-field or characteristic restriction is needed.
Input system	p. 3 writes a finite list of generators	line 42 permits a subset	A mild generalization, justified below under the target's explicit finiteness hypothesis.
Initial vectors	p. 6, Definition 1.5	lines 45–47	Same degree-truncated generator inclusion.
Closure	p. 6, Definition 1.5	lines 48–49	Same multiplication closure for every element already in the space.
Last-fall quantifier	p. 6, Definition 1.5	lines 51–52	Both quantify over every element of the generated ideal with budget max(d, deg f).
Individual entry degree	p. 9, Definition 2.5	lines 54–56	Same least membership budget and strict inequality defining a fall.
Existence of a fall	p. 9, Theorem 2.8	line 58	Explicit in both; it makes the fall-degree set nonempty.

Finiteness	p. 6, paragraph following Definition 1.5; used on p. 9	lines 27–31, 58	Paper supplies it from Gröbner-basis existence; target assumes it. This is an explicit conditional scope.
Greatest fall degree	pp. 9–10, Theorem 2.8	lines 60–73	Same ideal-wide maximum and both inequalities.

The paper defines auxiliary budgets at infinity and at −1. The target omits those auxiliary values and asks for a finite last-fall degree. They do not enter the finite maximum argument. The omission would matter for a claim to formalize all of Definition 1.5, but does not alter this conditional theorem. In particular, the paper's special V_{F,-1} = empty set should not be silently identified with the zero vector subspace.

Derivation checked in this review

This is a source-aware check, not the plan's blind re-derivation. Let I=(F), use nonnegative total degree with deg(0)=0, and let V_i have the stated generator and multiplication closure. Then V_i lies in I ∩ R_{≤i}, since that intersection is itself an admissible space. Also V_i ⊆ V_j for i≤j: V_j satisfies every defining obligation for V_i.

For every f∈I, finite d_F gives membership at max(d_F,deg f), so the set defining d_f is nonempty. The degree bound on V_i gives d_f ≥ deg f. No finiteness of the set of generators was used in these deductions. Thus the target's arbitrary-subset wording is a valid extension of this conditional argument, even though the paper's global setup names a finite system.

Write S={d_f : f∈I and deg f<d_f}. For any witness in S, if d_F<d_f, both d_F and deg f are less than d_f. Definition 1.5 would put f in a space with budget smaller than its minimum entry degree, a contradiction. Consequently S is bounded above by d_F. The fall hypothesis makes S nonempty, so this subset of natural numbers has a maximum mu. For any ideal element, either d_f=deg f, or d_f≤mu. Monotonicity gives membership in V_{max(mu,deg f)} in both cases. Hence mu is an admissible last-fall budget, giving d_F≤mu. Together the inequalities prove the target claim.

There is a local typo in the paper's first proof sentence on p. 9: it starts with f∈F, whereas its displayed supremum and the argument require f∈(F). This was checked in the PDF image, not inferred from OCR alone. Every generator already enters at its own degree, so restricting this step to generators cannot bound all ideal-element falls. The target's proof sketch uses the correct ideal-element reading. This source typo does not supply a counterexample to the theorem; the preceding argument repairs that quantifier explicitly.

Empty and no-fall controls

These are exact mathematical checks, not measured trials.

System in k[x]	Closure spaces	Last-fall degree	Fall-degree set	Theorem 2.8 applies?
F=empty	V_i={0} for all natural i	0	empty	No: fall hypothesis fails.
F={0}	Same spaces as empty F	0	empty	No.
F={x}	V_i=(x)∩R_{≤i}	0	empty	No.
F={1}	V_i=R_{≤i}	0	empty	No.
F={x^2,x^2+x}	V_0=V_1={0}; V_i=(x)∩R_{≤i} for i≥2	2	{2}	Yes.

For {x}, multiplying the generator by every polynomial of degree at most i−1 gives exactly the polynomials in (x) of degree at most i. Each nonzero ideal element therefore has d_f=deg f. For the positive control, subtraction produces x at budget 2, while budget 1 contains no nonzero generator and stays zero. Multiples of x fill the truncated ideal at every budget at least 2. Thus both hypotheses are jointly satisfiable: d_F=2 and d_x=2>1.

An empty set has no greatest element. Giving an empty supremum the value zero in an ordered type is a separate convention and cannot satisfy the membership part of a greatest-element assertion. The no-fall branch is instead covered by Lemma 2.7 (p. 9), which yields d_F=0. The paper does not explicitly spell out the zero-polynomial degree convention. The nonnegative convention used here makes d_0=deg(0)=0; alternatively, exclude zero from fall witnesses. Counting zero as a strict fall using deg(0)=-infinity would conflict with the paper's no-fall lemma and its homogeneous examples. The concrete Lean convention is the other reviewer's assignment.

Adjacent source qualifications for the coordinator

These are source observations, not a verdict on the measurement joint. On p. 4, M_d is the output after repeatedly reducing and adjoining permitted multiples of newly produced lower-degree rows until closure. Equation (1) on p. 6 refers to that object and a degree-compatible order. The phrase “after reduction” in producer-note line 27 does not itself spell out that iteration. Page 7 also expressly warns against stopping at one equality of adjacent truncated spaces; Example 2.4 illustrates the failure on pp. 8–9. The formal target's measurement prose at lines 13–16 therefore needs the separate review already assigned to T-1107.

The producer note's separate Theorem 2.9 line 35 omits the paper's explicit fall-existence condition (p. 10). This is an adjacent source transcription issue: for {x} its displayed maximum is empty unless a convention is added. It does not change the correctly conditional Theorem 2.8 row. No verdict is offered here on the note's other invariant or Weil-restriction claims.

Procedure, artifacts, and next action

The Conductor task T-1105 was claimed. The reviewer worked from the pushed plan in a separate sparse worktree and wrote only this source/ directory. The checkout was compared with the claim snapshot for both target and note; no differences were found. No sibling reports were opened. Repository worktree metadata was listed for orientation; that did not expose sibling report contents. The initial broad filename-only search was interrupted and returned no files. No machine-verification verdict is claimed.

review_attestation.yaml lists actual source and instruction reads; source_manifest.yaml binds the external PDF and local claim inputs. diagram.svg and report.typ are editable visual sources; report.pdf is the rendered source review. visual-inspection.json records

inspection. The canonical source note and target were checked as above. No canonical graph was changed: this source joint confirms the existing conditional theorem, and the reviewer's write scope excludes shared artifacts.

Next action: combine this source verdict with the independent Lean and measurement reports; preserve the no-fall branch and the explicit finiteness hypothesis when describing the conditional theorem. Any adjacent note correction should be additive and separately scoped.